

Algebra 1
Unit 5
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

1. When solving an equation, the goal is to isolate the variable.

Two students, Josephine and Jackie, were solving the equation $39 - 8n = -8(3 + 4n) + 3n$. Consider their work is shown below for questions 2–5.

Josephine's Work	Jackie's Work
$ \begin{array}{r} 39 - 8n = -8(3 + 4n) + 3n \\ \underline{-3n} \qquad \qquad \underline{-3n} \\ 39 - 11n = -8(3 + 4n) \\ 39 - 11n = -24 - 32n \\ \underline{+32n} \qquad \qquad \underline{+32n} \\ 39 + 21n = -24 \\ \underline{-39} \qquad \qquad \underline{-39} \\ 21n = -63 \\ \underline{21} \quad \underline{21} \\ n = -3 \end{array} $	$ \begin{array}{r} 39 - 8n = -8(3 + 4n) + 3n \\ \underline{-8} \qquad \qquad \underline{-8} \\ 4\frac{7}{8} + n = 3 + 4n + 3n \\ 4\frac{7}{8} + n = 3 + 7n \\ \underline{-3} \qquad \underline{-3} \\ 1\frac{7}{8} + n = 7n \\ \underline{-n} \quad \underline{-n} \\ \underline{1\frac{7}{8}} = 6n \\ \underline{6} \quad \underline{6} \\ \frac{5}{16} = n \end{array} $

2. Describe the different first steps the students used in solving the equation.
Josephine subtracted $3n$ and then distributed the -8 before subtracting again. Jackie divided everything by -8 to avoid distributing it, then she subtracted 3, then n , and finally divided by 6.
3. Which student made an error?
Jackie
4. Write a note to the student who made an error explaining the mistake they made.
When multiplying or dividing on both sides of an equal sign, always double check to make sure every term is being multiplied or divided.
5. Is -7 a solution to the equation $32 + 7x - 4 = 3x$? Justify your answer mathematically and using words.
Yes, the left-hand side equals the right-hand side when $x = -7$.
- $$\begin{array}{rcl}
 32 + 7x - 4 & = & 3x \\
 -7x & - & 7x \\
 \hline
 28 & - & 4x \\
 -4 & - & -4 \\
 -7 & = & x
 \end{array}$$
- $$\begin{array}{rcl}
 32 + 7(-7) - 4 & = & 3(-7) \\
 -21 & = & -21 \checkmark
 \end{array}$$

For questions 6–10, solve each equation and verify the solution using substitution.

6. $\frac{2m-5}{3} = 4m - 3$

$$3\left(\frac{2m-5}{3} = 4m - 3\right)$$

$$2m - 5 = 12m - 9$$

$$\frac{-2m}{-5} = \frac{-2m}{-5}$$

$$-5 = 10m - 9$$

$$\frac{+9}{+9} = \frac{+9}{+9}$$

$$\frac{4}{10} = \frac{10m}{10}$$

$$\frac{2}{5} = m$$

$$\frac{2}{5} = m$$

$$\frac{2\left(\frac{2}{5}\right) - 5}{3} = 4\left(\frac{2}{5}\right) - 3$$

$$\frac{-\frac{21}{5}}{3} = \frac{8}{5} - \frac{15}{5}$$

$$-\frac{7}{5} = -\frac{7}{5} \checkmark$$

7. $-11 - 2p = 6p + 5(p + 3)$

$$-11 - 2p = 6p + 5p + 15$$

$$\frac{+2p}{+2p} = \frac{+2p}{+2p}$$

$$-11 = 13p + 15$$

$$\frac{-15}{-15} = \frac{-15}{-15}$$

$$\frac{-26}{13} = \frac{13p}{13}$$

$$\frac{-2}{1} = p$$

$$-2 = p$$

$$-11 - 2(-2) = 6(-2) + 5((-2) + 3)$$

$$-11 + 4 = -12 - 10 + 15$$

$$-7 = -7 \checkmark$$

8. $\frac{2}{5} + 13k - 1 = 5k - \frac{3}{5}$

$$\frac{-5k}{-5k} = \frac{-5k}{-5k}$$

$$-\frac{3}{5} + 8k = -\frac{3}{5}$$

$$\frac{+3}{+3} = \frac{+3}{+3}$$

$$\frac{8k}{8} = \frac{0}{8}$$

$$\frac{k}{1} = \frac{0}{1}$$

$$k = 0$$

$$\frac{2}{5} + 13(0) - 1 = 5(0) - \frac{3}{5}$$

$$\frac{2}{5} - 1 = -\frac{3}{5}$$

$$-\frac{3}{5} = -\frac{3}{5} \checkmark$$

9. $3.1 + 6.8z - 2.4 = 0.3z - 4$

$$\frac{+4}{+4} = \frac{+4}{+4}$$

$$6.8z + 4.7 = 0.3z$$

$$\frac{-6.8z}{-6.8z} = \frac{-6.8z}{-6.8z}$$

$$\frac{4.7}{-6.5} = \frac{-6.5z}{-6.5}$$

$$-0.72308... = z \text{ or } z = -\frac{47}{65}$$

$$3.1 + 6.8\left(-\frac{47}{65}\right) - 2.4 = 0.3\left(-\frac{47}{65}\right) - 4$$

$$0.7 - \frac{1,598}{325} = -\frac{141}{650} - 4$$

$$-\frac{2,741}{650} = -\frac{2,741}{650} \checkmark \quad \text{or}$$

$$-4.21692... = -4.21692... \checkmark$$

$$10. \quad \frac{1}{2}x - 2(x + 1) = 3 + 1.5x + 1$$

$$\left(\frac{1}{2}x - 2(x + 1) = 3 + 1.5x + 1\right) 2$$

$$x - 4(x + 1) = 6 + 3x + 2$$

$$x - 4x - 4 = 3x + 8$$

$$\quad \quad \quad \begin{array}{r} +4 \qquad \qquad +4 \\ \hline \end{array}$$

$$-3x = 3x + 12$$

$$\begin{array}{r} -3x \quad -3x \\ \hline \end{array}$$

$$\begin{array}{r} -6x \quad 12 \\ \hline \end{array}$$

$$\begin{array}{r} -6 \quad -6 \\ \hline \end{array}$$

$$x = -2$$

$$\frac{1}{2}(-2) - 2(-2 + 1) = 3 + 1.5(-2) + 1$$

$$-1 + 2 = 4 - 3$$

$$1 = 1 \checkmark$$